Tarek Hamid

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My research develops **devices and methods for wearable physiological sensing** that make health monitoring personalized, robust, and equitable. I focus on (1) **context-aware signal processing** that measures and accounts for sensing confounders, such as skin optical properties and motion artifact, rather than treating them as noise; (2) **novel biomarker extraction**, including blood pressure, blood glucose, and nutritional metabolites, from physiological data; (3) **distributed and multi-modal sensing**, integrating signals across body sites, form factors, and clinical environments to build richer physiological representations than any single sensor can provide; and (4) **robust inference across monitoring contexts**, developing methods that maintain estimation quality as the sensing environment changes.

EDUCATION

University of Virginia, Charlottesville, VA *Jan 2024 - Present*
Ph.D. in Electrical and Computer Engineering
Dissertation: *Broad-Spectrum Multi-Wavelength PPG Algorithms for Personalized and Robust Wearable Physiological Monitoring*
Advisor: Prof. Amanda Watson

University of Pennsylvania, Philadelphia, PA *Dec 2023*
M.S. in Computer Science (50% Completed)

Johns Hopkins University, Baltimore, MD *Dec 2019*
M.S. in Biomedical Engineering

The College of New Jersey, Ewing, NJ *May 2017*
B.S. in Biomedical Engineering

HONORS AND AWARDS

Google PhD Fellowship Finalist *Apr 2026*
National finalist for the Google PhD Fellowship ($170,000 support over two years)

Runner-up for Best Paper: IEEE Pervasive Computing *Apr 2026*
Characterization and Feasibility of Wearable Spectroscopic Tracking of Nutrition Biomarkers was selected as runner-up for best paper of IEEE Pervasive Computing 2025.

Best Talk: SandboxAQ 2026 Conference *Mar 2026*
Award of $300 for best talk at the 2026 SandboxAQ Conference.

SandboxAQ Global Travel Award *Aug 2025*
Award of $5,000 to cover travel expenses to present at UbiComp'2025 in Helsinki, Finland.

UVA Precision Health Initiative Grant *Feb 2025*
Award of $4,000 to cover travel expenses to present at ICASSP'25 in Hyderabad, India.

Runner-up for Best Paper Award, IEEE BSN 2024 *Oct 2024*
Presented Raproto: An Open-Source Platform for Rapid Prototyping with Wearable Devices at IEEE Body Sensor Networks 2024.

TCNJ Merit Scholarship *Aug 2013 - May 2017*
Award of $32,000 for educational expenses.

PUBLICATIONS

1. *DermaGlow: Objective Quantification of Melanin, Erythema and Skin-tone Using Wearable Optical Spectroscopy*
   Tarek Hamid, Patricia Flores, Jane Byun, Elizabeth C. Courtney, Kyle C. Quinn, Amanda Watson
   Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT).
   US Patent #10885-083US1

2. *Characterization and Feasibility of Wearable Spectroscopic Tracking of Nutrition Biomarkers*
   Tarek Hamid, Elizabeth Courtney, Patricia Flores, Jane Byun, Afsaneh Doryab, Sibylle Kranz, Amanda Watson
   Pervasive'25: IEEE Pervasive Computing: Special Issue on Bio Sensing
   Featured Paper, 2025 Best Paper Runner-Up

3. *A Multi-Wavelength Optical Sensing Framework for Calibration-Free Wearable Blood Pressure Monitoring*
   Tarek Hamid, Patricia Flores, Jane Byun, Xi Chen, Haoran Zhang, Kyle Quinn, Amanda Watson
   International Conference on Acoustics, Speech, and Signal Processing (ICASSP) 2025.
   US Patent #10885-090US1

4. *Raproto: An Open-Source Platform for Rapid Prototyping with Wearable Devices*
   Tarek Hamid, Kimberly Helm, Hyonyoung Choi, Jean Park, Claire Kendell, Stephanie Cummings, Steve Messe, Stefanie Modri, Insup Lee, Amanda Watson, James Weimer
   Proc. IEEE-EMBS Int. Conf. on Body Sensor Networks (IEEE BSN), 2024.
   Best Paper Runner-Up

5. *Wearable Sensing for Measuring Skin-tone, Melanin, and Erythema*
   Tarek Hamid, Anush Lingamoorthy, Kyle Quinn, and Amanda Watson
   Proc. IEEE-EMBS Int. Conf. on Body Sensor Networks (IEEE BSN), 2024.

6. *Using Decision Tree Classifier to Increase Screening Test Sensitivity for the Prediction of ACL Retear*
   Tanishik Govil, **Tarek Hamid**, Kimberly Helm, Elliot Greenberg, Kevin Landrum, J. Todd R. Lawrence, Theodore J. Ganley, Amanda Watson
   UBICOMP'24: Adjunct Proceedings of the 2024 ACM International Joint Conference on Pervasive and Ubiquitous Computing, Melbourne, Australia 2024

7. *SpectraVue - An Interactive Web Application Enabling Rapid Data Visualization and Analysis for Wearable Spectroscopy Research*
   Tarek Hamid, Insup Lee, Amanda Watson
   UBICOMP'23: Adjunct Proceedings of the 2023 ACM International Joint Conference on Pervasive and Ubiquitous Computing, Cancun, Mexico 2023

8. *Alleviation of Arthritic Symptoms through Thermal Therapy*
   Tarek Hamid, Steven Ayala, Aakash Trivedi, Avi Shah
   In Proceedings of the 2017 Northeast Biomedical Engineering Conference (NEBEC). Newark, NJ: IEEE

9. *(In review) Chronological Age Estimation using Multi-Wavelength PPG*
   Tarek Hamid, Anush Lingamoorthy, Amanda Watson.

10. *(In review) Accelerometer-Free Pulse Recovery using Multi-Wavelength PPG*
    Tarek Hamid, Patricia Flores, Amanda Watson.

11. *(In review) Contact Pressure Estimation using Multi-Wavelength PPG*

Tarek Hamid, Patricia Flores, Amanda Watson.

12. *(In review) Glucose Estimation using Multi-Wavelength PPG*
    Anush Lingamoorthy, Abhishek Murtha, **Tarek Hamid**, Kyle Quinn, Nagarajan Kandasamy, Amanda Watson.

13. *(In review) Lactate Estimation using Multi-Wavelength PPG*
    Haoran Zhang, Patricia Flores, Michael Clark, Yuxuan Liu, Elizabeth C. Courtney, Xi Chen, **Tarek Hamid**, Robert Theile, Emily Grammar, Kyle C. Quinn, Amanda Watson

INVITED TALKS (EXCLUDES CONFERENCE PRESENTATIONS)

Navigating the Industry as a New Graduate *November 2024*
Duke Industry Series for AI Master of Engineering Students

PATENTS

1. *Objective quantification of melanin, erythema and skin-tone using wearable optical spectroscopy*
   Tarek Hamid, Amanda Watson, Kyle Quinn
   US Patent #10885-083US1 (Pending)

2. *System and Method for Multi-Wavelength Optical Sensing Framework for Calibration-Free Wearable Blood Pressure Monitoring*
   Tarek Hamid, Amanda Watson
   US Patent #10885-090US1 (Pending)

SOFTWARE ARTIFACTS

1. SpectraVue - rapid data visualization and analysis for Lumos
   https://github.com/tarek-hamid/SpectraVue

2. Lumos: An Open-Source Device for Wearable Spectroscopy Research
   https://github.com/LumosIMWUT/Lumos

3. *Raproto: An Open-Source Platform for Rapid Prototyping of Wearable Medical Devices*
   https://github.com/weimerj/Raproto-WearOS
   https://github.com/weimerj/Raproto-Arduino

4. *Digital Biomarker Discovery Pipeline GUI*
   https://github.com/DigitalBiomarkerDiscoveryPipeline/DBDP-GUI

RESEARCH EXPERIENCE

PRECISE Center *2020 - 2022*
University of Pennsylvania

- Developed and integrated IoT-based health monitoring systems using ThingsBoard and VitalCore, to enable real-time physiological monitoring and data processing for devices deployed in clinical trials.
- Designed and developed the Raproto mobile application, a mobile application enabling rapid prototyping of Android smartwatches for data collection and monitoring.
- Performed signal processing and data analysis for the Lumos glucose project, developing pipelines to extract meaningful physiological features from optical sensor data.

Big Ideas Lab *2021 - 2022*
Duke University

- Built an interactive web-based dashboard (DBDP-GUI) that enabled clinicians and researchers to define and explore digital biomarkers from wearables data.

ENTREPRENEURIAL ACTIVITIES

Co-founder *Jan 2026 - present*
OxyVera
Developing the first skin-tone invariant pulse oximeter.

MENTORED STUDENTS

1. **Patricia Flores**, Undergrad Research on *Development of Infrared Wearable Optical Sensors for Biomarker Monitoring* *2024-present*
2. **Tanishk Govil**, Undergrad Research on *Noninvasive Continuous Blood Pressure* *2024-present*
3. **Tanishk Govil**, Undergrad Research on *ACL Re-tear Prediction* *2023-2024*
4. **Xi Chen**, Masters Research on *Robust Pulse Rate Extraction across Skin Tones Using Multi-Wavelength PPG* *2024-2025*
5. **Wilson Beckman**, Post-Undergraduate Mentorship: *Now Machine Learning Data Scientist at Ascension* *2021-2022*
6. **Celine Harris**, TCNJ Engineering Mentorship Program: *Now Senior Associate at Grant Thornton* *2020-2021*
7. **Daniel Young**, Post-Undergraduate Mentorship: *Now MS in Applied Economics at Boston University* *2020-2021*
8. **Harish Erra**, TCNJ Engineering Mentorship Program: *Now Account Executive at Fingerpaint Medical* *2018-2019*

PROFESSIONAL SERVICE AND AFFILIATIONS

Workshop/Conference Organization

1. IEEE International Conference on Healthcare Informatics *2026*
   TRUST-AI 2026: Trustworthy AI for IoMT *Technical Program Committee*

Journal and Conference Reviews

1. *International Conference on Acoustics, Speech, and Signal Processing (ICASSP)* *2025, 2026, 2027*
2. *ACM Conference on Bioinformatics, Computational Biology, and Health Informatics* *2026*
3. *IEEE Body Sensor Networks (BSN)* *2025, 2026*
4. *IEEE Engineering in Medicine and Biology Society (EMBC)* *2024, 2025, 2026*
5. *ACM Health* *2025, 2026*
6. *Elsevier Smart Health* *2024, 2025*

Professional Affiliations

1. *IEEE EMBS: Technical Community on Wearable Biomedical Sensors and Systems (WBSS)*
2. *IEEE*

3. *ACM*
4. *SIGCHI*

TEACHING EXPERIENCE

Teaching Assistant *Fall 2025*
ECE 4501/6501: Sensors and Ubiquitous Computing
Department of Electrical and Computer Engineering, University of Virginia

INDUSTRY EXPERIENCE

Sotera Digital Health Jul 2022 - Dec 2023
Biomedical Algorithm Engineer *Remote*

Designed and implemented signal processing and data science algorithms in Python for a next-gen hospital wearable, extracting vital signs such as heart rate, SpO2, and blood pressure from ECG, PPG, and SCG sensor data.

JPMorgan Chase Jun 2020 – July 2022
Software Engineer *New York, NY*

Developed algorithms and internal customer-facing applications to report on the stability of new code changes to the Chase Consumer application using Python, Java, Angular, and TypeScript.

Department of Defense Oct 2019 – June 2020
Electrical Engineer *Picatinny Arsenal, NJ*

Designed custom hardware solutions for military vehicles using Altium.

Johnson & Johnson Jan 2018 – Oct 2019
Scientist *Skillman, NJ*

Led R&D lifecycle management activities for Class I and II consumer medical devices in the North American region.